

LINCOLN COUNTY AP, NC, USA

WMO: 722128

Lat: 35.483N

Lon: 81.161W

Elev: 875

StdP: 14.24

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 53899

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	22.4	-8.5	3.6	38.7	-0.2	5.6	35.8	18.5	49.7	16.6	49.9	1.0	300	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	94.2	72.7	91.6	72.2	90.2	72.0	78.3	86.2	76.7	85.3	75.2	83.5	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	138.4	80.9	74.4	132.6	79.5	72.9	126.3	78.0	42.4	87.7	40.6	84.9	39.4	82.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.4	12.4	10.6	DB	9.3	97.3	5.2	2.7	5.5	99.2	2.5	100.8	-0.5	102.3	-4.3	104.2
			WB	8.1	78.5	4.5	2.6	4.9	80.4	2.2	81.9	-0.3	83.3	-3.6	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	40.5	44.4	50.7	59.5	67.7	75.8	78.6	77.7	72.3	60.3	49.1	43.9	(6)
	DBStd	15.42	9.87	9.10	9.41	7.76	6.20	4.14	3.66	4.20	5.38	7.84	8.19	8.55	(7)
	HDD50	1007	322	202	112	15	0	0	0	0	0	11	115	230	(8)
	HDD65	3366	760	578	450	197	48	1	0	0	11	187	480	654	(9)
	CDD50	4699	28	45	134	298	548	775	886	858	668	330	87	42	(10)
	CDD65	1583	0	1	7	31	131	326	421	393	229	41	2	1	(11)
	CDH74	14637	0	11	112	420	1287	3022	3976	3509	1899	375	22	4	(12)
	CDH80	6041	0	0	15	80	399	1300	1861	1574	743	68	1	0	(13)
(14) Wind	WSAvg	3.0	3.4	3.9	4.3	4.4	3.0	2.5	2.2	2.0	2.5	2.6	2.7	2.8	(14)
(15) Precipitation	PrecAvg	43.5	3.9	3.2	4.3	3.4	3.3	4.4	3.7	4.3	3.6	3.4	2.9	3.3	(15)
	PrecMax	62.9	6.3	5.8	8.8	9.1	10.2	8.7	8.6	9.1	7.7	12.8	6.0	5.8	(16)
	PrecMin	30.6	1.1	1.3	1.3	0.6	1.4	1.4	1.9	0.6	0.4	0.0	0.7	1.4	(17)
	PrecStd	8.0	1.5	1.3	1.9	2.0	2.1	2.2	1.5	2.6	1.7	3.0	1.5	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	75.4	82.1	85.7	90.6	96.4	96.9	98.6	93.1	85.2	77.2	72.5	(19)
		MCWB	58.1	61.0	59.5	61.9	68.5	72.8	73.4	70.2	67.4	67.5	60.8	60.3	(20)
	2%	DB	65.7	70.1	77.1	82.0	87.9	92.8	93.7	93.3	90.6	81.7	72.4	67.2	(21)
		MCWB	56.5	56.8	58.6	60.6	68.1	71.7	74.1	72.8	68.7	65.4	59.1	60.5	(22)
	5%	DB	62.7	65.3	72.8	79.4	84.3	90.3	91.3	90.8	88.1	78.9	69.1	63.3	(23)
		MCWB	55.1	55.1	56.7	60.2	67.0	71.7	73.7	73.0	69.5	64.1	58.7	57.1	(24)
	10%	DB	57.3	61.3	68.2	75.1	81.5	87.8	89.8	88.4	84.3	74.6	64.9	59.4	(25)
		MCWB	50.2	50.8	54.7	58.8	65.9	71.1	73.3	73.2	69.3	64.0	56.4	52.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	64.1	65.1	68.8	74.0	77.8	80.0	81.6	78.0	73.4	67.6	66.4	(27)
		MCDB	64.6	70.2	71.5	75.2	82.7	86.0	87.5	86.4	82.9	76.3	70.8	69.5	(28)
	2%	WB	59.9	61.6	62.4	66.6	72.0	75.9	78.0	78.9	74.5	71.8	64.2	63.1	(29)
		MCDB	64.3	65.8	70.3	74.3	80.3	84.8	87.5	85.5	80.9	76.0	68.0	65.2	(30)
	5%	WB	56.1	57.9	60.1	64.5	70.4	74.6	76.6	76.7	73.0	68.8	61.5	59.2	(31)
		MCDB	60.6	64.8	69.3	71.6	78.5	83.7	86.2	84.7	79.9	73.8	65.9	62.1	(32)
	10%	WB	52.2	53.1	57.6	62.8	68.9	73.3	75.2	75.1	71.7	66.4	58.7	54.3	(33)
		MCDB	56.5	58.0	65.7	70.3	76.9	82.4	84.3	82.6	78.3	72.1	63.5	57.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	22.4	24.0	24.7	23.2	21.7	21.0	20.3	20.9	23.8	24.9	21.0	(35)
		MCDBR	26.2	27.5	28.7	29.1	27.2	26.3	23.6	23.8	26.6	27.2	29.1	24.8	(36)
		MCWBR	17.1	16.8	13.6	12.2	10.3	8.9	7.4	7.8	8.6	12.6	16.8	16.6	(37)
	5% WB	MCDBR	20.9	24.0	23.8	21.8	21.7	22.3	21.1	19.9	20.1	17.9	20.2	19.4	(38)
		MCWBR	17.3	17.3	14.0	11.9	9.5	8.7	8.1	8.5	8.8	10.2	15.1	16.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.311	0.320	0.357	0.400	0.453	0.481	0.529	0.516	0.421	0.377	0.328	0.312	(40)
	taud		2.495	2.464	2.398	2.278	2.163	2.122	2.015	2.032	2.285	2.388	2.485	2.523	(41)
	Ebn at Noon		283	293	290	281	266	258	244	245	265	268	274	276	(42)
	Edh at Noon		26	30	36	42	48	50	55	53	39	32	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		829	1037	1371	1708	1901	2023	1888	1735	1469	1210	916	710	(44)
	RadStd		50	120	97	132	154	145	129	83	133	149	76	67	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+292	+196

Nomenclature: See separate page